TRACE-RPG: Auditable Proposal-to-Commit Boundaries and Source-Bound Read-Only Ontology Observation in Interactive Games

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***Abstract*—TRACE-RPG combines external deterministic authorization, typed-error repair, unchanged fallback, pre-commit revalidation, semantic replay, and assignment-complete accounting; its read-only bilingual ontology view binds source declarations to public observations without mutation authority.**

We report bounded conformance and exploratory observations across three active surfaces: (1) an operator-assisted Codex dashboard case (n=1, 0 independent repeats) under `gpt-6.1-sol` with repaired mobile geometry, 62/62 checks, a Godot Web build, and real-input denial/disclosure/redisplay; (2) a fixed-game native pilot on one authored quest (seed 20261024; 8 assigned, 7 observed) yielding 5 system-goal successes, 1 pre-response harness-serializer failure (not rerun), and 2 `gpt-6.1-sol` ontology timeouts, alongside 23 host invocations, 0 repairs, 0/16 unsafe transitions, 7 valid semantic-link replays, and 6 valid fresh-engine reexecutions (1 `NOT_RUN`, 1 unobserved); and (3) a synthetic v5 paired-selector collection (20/20 responses received) where 8/20 passed the strict gate, 7/20 reached the specified goal, and 10/20 failed schema parsing. Provider identity/revision/cost/request count, pure inference latency, and the dashboard seed remain unavailable, and the native pilot carries a controller-v3/v4 revision confound (row 1 vs. rows 2–8). Accordingly, these results establish bounded integration and accounting conformance rather than confirmatory questions A/C, ontology superiority, model ranking, or player benefit.

***Index Terms*—game AI, interactive narrative, symbolic validation, constrained generation, reproducibility**

I. Introduction

GENERATED events may request absent items, unknown facts, forbidden disclosures, unauthorized effects, or regressions. Plausibility and grounding [1], [2] do not authorize mutation: **only TRACE-RPG's external deterministic gate authorizes mutation** (Fig. 1).

Our contribution is a scoped proposal-to-commit integration and its evidence-accounting design: repair information can change the candidate, but only revalidation against the same authored constraints can authorize a state change; observed checks and replay have explicitly separate evidential limits. C1–C5 organize these contracts and their evidence, rather than claiming new constituent algorithms or demonstrated model advantage.

Table I scopes **C1—contracts**, **C2—validation/repair**, **C3—zero-regression audit binding**, **C4—assignment accounting**, and **C5—guided repair** to their evidence and inference ceilings.

TABLE I
Contributions, Prior Topics, Evidence Lanes, and Inference Ceilings (Generated from the Structured Matrices)

ID	Contribution	Prior topics (refs)	Lanes	Inference ceiling
C1	Trust-boundary contracts	T1, T2, T3 (17)	E1, ENG1	encoded-field contract conformance
C2	Validate–repair–commit controller	T1, T8 (14)	E1, ENG1	mechanism behavior on frozen cases
C3	Audit-linked evidence layer	T4, T7, T8 (6)	E1, ENG1	integrity/replay on named fixtures
C4	Assignment-complete harness	T1, T4, T6, T7 (20)	E1	exact designed-case accounting
C5	Counterexample-guided operator ρ	T1, T8, T9 (8)	E1	frozen offline repairability classes

T1 = grounded game worlds; T2 = retrieval/memory agents; T3 = structured generation; T4 = environments/benchmarks; T5 = player experience; T6 = human/LLM evaluation; T7 = statistics/reproducibility; T8 = planning/runtime interposition; T9 = feedback repair. Lanes are defined in Table V. All rows are authored-fixture verified; none supports an efficacy claim.

II. Related Work and Research Gap

Grounding/memory/planning/role [1], [2], [3], [4] and player/dialogue work [5], [6], [7], [8] distinguish believability from admissibility; affect/engagement [9], [10], [11] remains future, unimplemented, and non-authoritative. The design question addressed here is how to keep mutation authority, repair information, and replay evidence distinct within a source-bound proposal-to-commit path; this is not a claim that prior work lacks these components.

Benchmarks and automated play-testing span state, embodiment, content, process, social interaction, and long-horizon play [12], [13], [14], [15], [16], [17], [18], [19], [20]; NCP-Bench tests narrative commitment/fact consistency, not player outcomes [21]. Structural/schema, affordance, and rule methods [22], [23], [24], [25], [26], [27], [28], [29], [30], [31], [32], [33] motivate constraints, not superiority; retrieval/memory supplies context without commit authority [34], [35], [36].

Reproducibility/evaluation motivates frozen artifacts, retained failures, construct/rater/bias controls, and justified margins/random effects [37], [38], [39], [40], [41], [42], [43], [44]; resource accounting is motivated by energy and carbon reporting work [45]. Replay/leakage checks support denominators, not efficacy [46], [47]. C1–C5 combine enforcement, state/transaction management, feedback repair, action

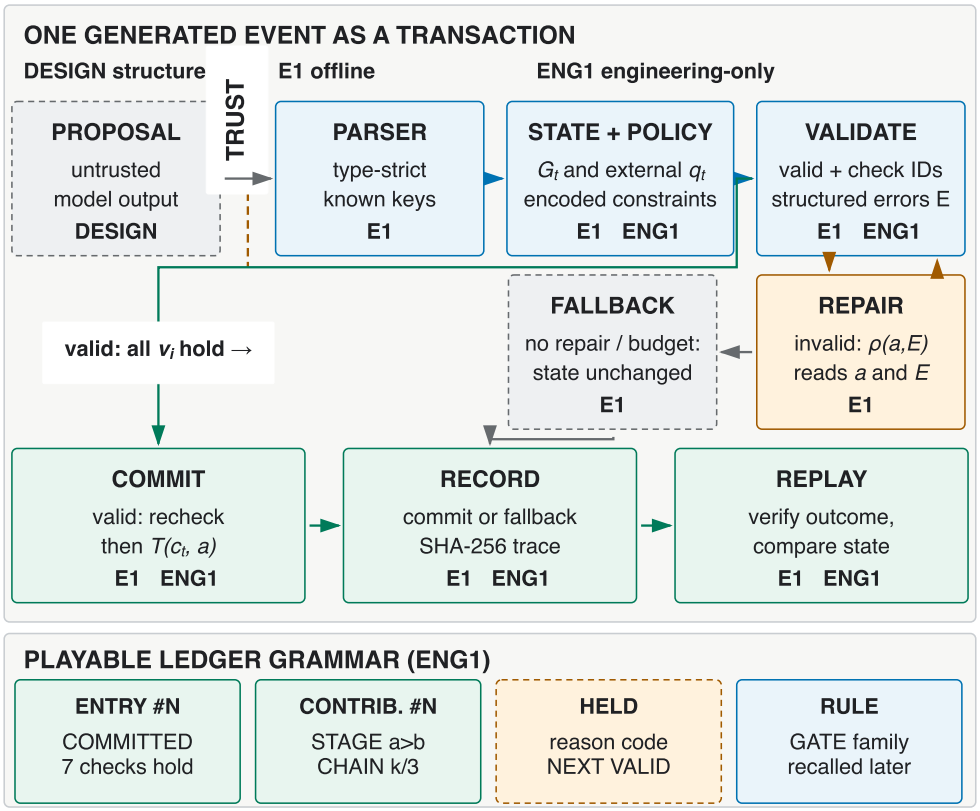


Fig. 1. Policy/c_t gates proposals; errors feed bounded ρ or unchanged fallback. Passing candidates are rechecked, committed, recorded, and replayed. Historical cohorts/legacy ENG1 remain archived, not active model evidence. Below: committed-snapshot ledger grammar.

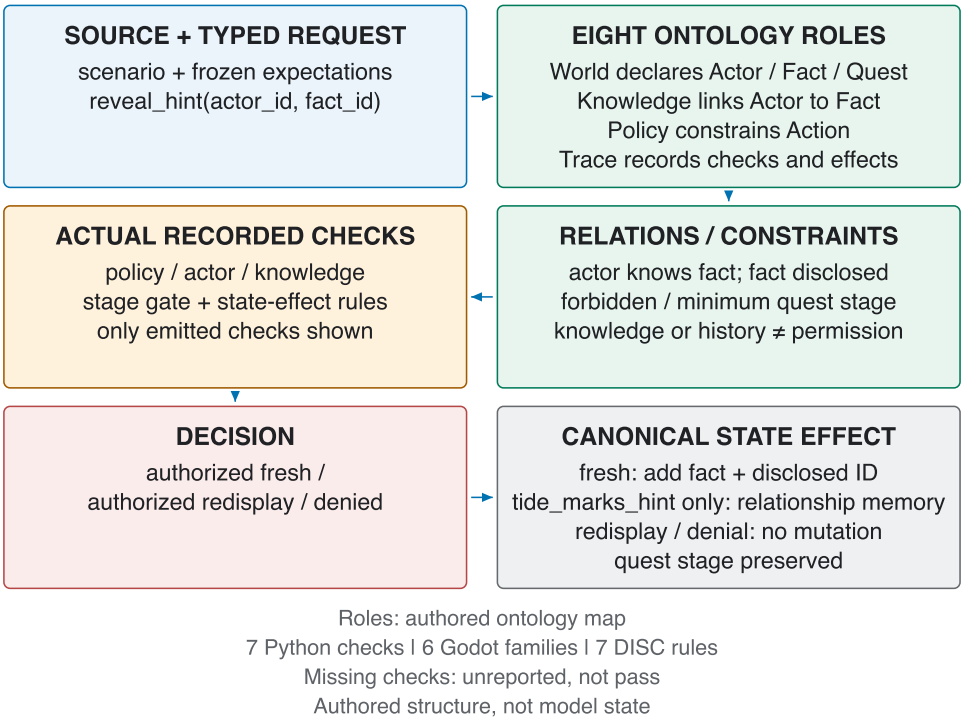


Fig. 2. Eight-role ontology structure (Table III) links source fields to checks and observed decisions. Only authorized transitions change canonical state; the diagram is neither live capture nor hidden-state evidence.

interposition, preconditions/effects, abort, social/belief-aware planning, and replay [27], [28], [29], [48], [46], [49], [50], [51], [52], [53], [54]; no invention/firstness claim.

III. Problem Definition and Guarantee Boundary

$c_t = (G_t, q_t)$ separates the typed game snapshot G_t from externally authored action-policy/disclosure/quest-stage constraints q_t. In Python these are carried together by `WorldState`: locations, reachable locations, object locations, inventory, facts, NPC knowledge, and quest stage describe the encoded state; action policies and forbidden disclosures constrain it. An `ActionPolicy` specifies required preconditions, allowed and required effects, and allowed quest-stage targets. This is a conceptual separation, not a claim that Python stores two independent objects. Predicates neither read $h_{<t}$ nor reconstruct `WorldState`; transitions update G_t only, excluding policy rotation. Retrieval, scores, rationales, memory, and future affect lack authority.

After strict parsing of the untrusted proposal a_t, the validator evaluates seven deterministic predicates and returns `ValidationResult(valid, errors, checks_run)`, which we model as

$$\begin{aligned} V(c_t, a_t) &= (\text{valid}, E_t, C_{\text{run}}), \\ \boldsymbol{v} &= (v_{\text{policy}}, v_{\text{pre}}, v_{\text{reach}}, v_{\text{know}}, \\ &\qquad v_{\text{disc}}, v_{\text{elig}}, v_{\text{stage}}) \in \{0,1\}^7, \end{aligned} \tag{1}$$

where E_t denotes structured errors (with code, entity, reason, and repair hint), C_{run} lists the seven executed check identifiers, and $\text{valid} \equiv \bigwedge_{i=1}^{7}(v_i = 1)$ summarizes the seven latent predicate outcomes (policy, precondition, reachability, NPC knowledge, forbidden disclosure, stage eligibility, and stage monotonicity). Because the runtime contract returns the aggregated `valid` flag and executed check identifiers rather than seven individual Booleans, an absent per-check trace or omitted field is treated as unobserved rather than passing or failing. The following equation describes a single-current-candidate gate, not the whole transaction for the original proposal. Here a_t is the current candidate, possibly changed by repair, and $\boldsymbol{v}$ denotes that candidate's checks against unchanged c_t. A rejected attempt leaves c_t unchanged while control continues through bounded repair or fallback; only passing same-policy revalidation authorizes mutation:

$$c_{t+1} = \begin{cases} T(c_t, a_t), & \bigwedge_{i=1}^{7}(v_i = 1), \\ c_t, & \text{otherwise.} \end{cases} \tag{2}$$

The operational path is proposal → deterministic validation → observable validity, check identifiers and structured errors → optional bounded repair → revalidation against the same state → commit or unchanged fallback. A valid candidate is checked again immediately before application; exhausted repair or no repairer leaves the state unchanged. The guided repair operator described below uses only the proposal and errors; the general Python repair callback nevertheless receives `WorldState`, so the restricted-input claim does not apply to every possible callback. For a controlled comparison, deterministic and model-based detailed repair must receive the same proposal and detailed errors, while the same model's generic-error condition receives a generic signal; no comparison outcome is implied by this design.

TABLE II
Encoded Validator Boundary

Predicate	Rejects	On failure
Action policy	unknown action/type; omitted required preconditions/effects; disallowed effects or unauthorized quest-stage effects	unchanged
Precondition	absent/false requirement	unchanged
Reachability	inaccessible object	unchanged
NPC knowledge	undeclared known fact	unchanged
Disclosure	forbidden fact	unchanged
Quest stage (`quest_stage`)	required stage above current	unchanged
Quest-stage monotonicity (`quest_stage_monotonicity`)	regressive stage	unchanged

Fixtures test invariants, not theorems. **I1:** validation exhaustion or pre-commit proposal/controller failure preserves state. **I2:** returning callbacks and $K \in \mathbb{Z}_{\geq 0}$ bound attempts to $K + 1$ with pre-commit revalidation; without deadlines, wall-clock termination is not guaranteed. **I3:** accepted `CandidateAction.effects` stay within `ActionPolicy.allowed_effects`, including authorized quest-stage targets. **I4:** given the recorded initial snapshot, a complete ordered transition record, and the frozen schema and deterministic transition version, replay can reconstruct the recorded terminal structured state. This is conditional reconstruction, not authentication, semantic correctness, or reproduction of model generation.

Assumptions: correct policy/extraction, deterministic versions, returning callbacks, single-process ownership. Replay checks structured transitions, not repair provenance, prompts, stochastic generation, or undeclared natural-language implications.

IV. TRACE-RPG Architecture and Algorithms

A. Trust Boundary and Contracts

External models/adapters propose; parser rejects ambiguous types/unknown keys; independent policy defines preconditions/effects. Versioned bridge carries observations, proposals, validations, rejections, fallbacks, and commits (Fig. 1); unknown-key parity proves neither parser equivalence nor portability. Python has exactly seven deterministic checks: `action_policy`, `precondition`, `object_reachability`, `npc_knowledge`, `forbidden_disclosure`, `quest_stage`, and `quest_stage_monotonicity`. Godot's six state-relative families and seven DISC rules are distinct, non-isomorphic labels; its `QUEST STAGE` display spans two Python quest checks.

The ontology role map binds source declarations to frozen expectations. Table III's `reveal_hint(actor_id, fact_id)` separates view, repair, and authority: the view exposes

TABLE III
EIGHT ONTOLOGY ROLES: READ-ONLY SOURCE BINDINGS, NOT LIVE INFERENCE OR AUTHORITY

Role	Source declaration or relationship example
World	Initial object locations place the signal lens in the lamp store.
Actor	Captain Mira is the sole eligible actor.
Knowledge	Actor knowledge refers to canonical Fact IDs, not permission.
Fact	Initial facts include `dock_saved` and `lighthouse_dark`; `keeper_betrayal` is an ID, not payload.
Policy	Disclosure policy forbids `keeper_betrayal`; `tide_marks_hint` needs minimum quest stage 2.
Quest	The initial quest is at stage 0 with flag `dock_saved`; disclosure preserves stage.
Action	`reveal_hint(actor_id,fact_id)` adds Fact/disclosed ID; only tide hint adds relationship memory. Fresh revision $+1$; redisplay unchanged.
Trace	Prior and next state hashes, rule and template identifiers, and case ID bind packet/version to evidence; absent fields cannot establish PASS.

`tide_marks_hint`/stage-2 policy; ρ sees only a_t, E_t, may insert a named predicate, and cannot read `WorldState`. Godot commits only all-seven-pass candidates; fresh disclosure increments revision $+1$; redisplay does not.

1) Zero-regression audit binding (C3): The eight-role contract binds source declarations, executed checks, decisions, state effects, and replay receipts without granting the read-only observer mutation authority. Under the same frozen NPC-DISC-1 rules, `general_rule` and `ontology_gate` agree on all 60/60 case outcomes, with zero observed expectation, state, display, or replay mismatches in either arm (Tables X–XII). This is a bounded zero-regression audit binding over declaration-derived fixtures, not a held-out safety or ontology-superiority result. The proposed TR01–TR06 rubric remains `PROPOSED_NOT_VALIDATED`; no measured traceability score or comparative traceability gain is reported.

B. Fixed-goal NPC integration

Under `npc-goal-recovery/1.1.0`, Codex proposes intent, Python checks public claims/plans, and only Godot writes state. The fixed harbor-restoration/authorized-tide-lead goal is not model-editable. Bindings cover session, epoch, request, scenario, NPC, goal, version, policy/config hashes, revision/state hash, deadline/budget, and issued action. Service: `127.0.0.1:8768`; browser origins: `localhost`, `127.0.0.1:8877`.

1) NPC-DISC-1 disclosure contract and authority boundary: NPC-DISC-1's 60 declaration-derived `reveal_hint` cases do not rerun/enlarge historical `verified-run-006`. Generated results/Tables X–XII define strata, bindings, and input-to-check mappings; Godot rechecks service bindings before mutation.

C. Validate–Repair–Commit

In Fig. 3's dependency-free pseudocode, the repair budget is $K \in \mathbb{Z}_{\geq 0}$ and Record appends in memory; persistence follows terminal consistency. (ok, E) abbreviates V with $\mathrm{ok} := \bigwedge_{i=1}^{7}(v_i = 1)$; $V(c,a).\mathrm{ok}$ revalidates.

1: $a \leftarrow \mathrm{Propose}(c)$; $j \leftarrow 0$
2: **repeat**
3: $(\mathrm{ok}, E) \leftarrow V(c,a)$; $\mathrm{Record}(a, E)$
4: **if** ok **then**
5: **assert** $V(c,a).\mathrm{ok}$; **return** $\mathrm{Commit}(c,a)$
6: **if** $j = K$ **then return** $\mathrm{Fallback}(c)$
7: $a \leftarrow \mathrm{Repair}(c,a,E)$; $j \leftarrow j+1$
8: **until terminal**

Fig. 3. Validate–repair–commit. At most $K+1$ candidate attempts are recorded; success defensively validates before mutation.

TABLE IV
PER-ERROR-CLASS EDITS OF THE GUIDED OPERATOR ρ

Validator error class	Edit $\delta(e)$ on the candidate
Policy precondition omission	insert the named predicate
Policy effect omission	insert the named effect
Policy effect violation	drop the named effect
Unsatisfied precondition	drop the named predicate
Unauthorized stage mutation	clear `quest_stage_effect`
Unknown action type	no-op (no registered alternative)
Unreachable object; knowledge, disclosure, and stage classes	no-op (candidate edit would launder or needs state)

Repair cannot mutate state; exhaustion preserves it. Typed `counterexample_guided_repair` reads only candidate a/errors E (its sole state-derived input), never `WorldState`:

$$\rho(a, E) = a \oplus \bigcup_{e \in E} \delta(e), \tag{3}$$

Table IV's $\oplus$ applies deterministic $\delta(e)$: one field/error entity, at most $|E|$/attempt; conflicting requirements/rejections leave fields unchanged. Fixed points require identical E, not convergence. The nondeployable state-reading oracle ignores errors, reconstructs preconditions/effects and both quest-stage fields from state, and never launders violations by deleting required objects/used facts/disclosed facts.

D. Integrity, Semantic Replay, and Accounting

Unkeyed SHA-256 detects accidental/tested changes, not identity or joint rewriting. Replay checks prior-state/candidate/validation, invalid nonfinal attempts, recomputed final state, and JSONL continuity, never repair. Request→decision→result and first-action/one-receipt bindings do not prevent duplicate IDs or impersonation. Terminal classes are commit, symbolic fallback, adapter failure, and controller failure (state/assignment preserved; no completed proposal trace). Pre-execution run/arm/scenario/seed/model/revision keys plus configuration/input/prior-state hashes bind assignments; aggregation rejects duplicate/missing/unexpected keys and separates failures.

V. EVIDENCE LANES AND BOUNDARIES

A. *E1* Offline Conformance Design

PQ1–PQ5 cover immutability, repair, checksum/semantic/linkage/continuity faults, rollback, fail-closed assignment/adapter/manifest accounting, and unknown-key

TABLE V
EVIDENCE LANES AT A GLANCE: DESIGN, UNIT, COMPARISON, AND CEILING (GENERATED FROM THE EXPERIMENT MATRIX)

Lane	Design	Unit / budget	Comparison	Ceiling
E1	Offline conformance: authored frozen fixtures, one world	13 gate; 12 repair/arm; 10 faults; 7 adapter; 3 guards	reject vs blind vs ρ vs oracle	mechanism conformance
E3	KG/ontology simulation: closed-world typed-link holdout	43 nodes; 106 ref. edges; 24 typed; 210 scores	degree baseline vs 6 fixed strategies	construction result only
ENG1	Engineering-only Godot/Web: engine-local policy mirror, no participants	4 fixtures; 52 checks; 8 smoke; 5 archetypes	authored fixture paths	engineering-only conformance

Active lanes are non-interchangeable with different denominators; the historical E2 receipts are excluded from active generation and ENG1 is engineering-only. No active lane establishes population efficacy, model ranking, player experience, retrieval benefit, or commercial-engine performance.

TABLE VI
SIMULATION-ONLY PROPOSAL-STRATEGY EVALUATION ON A FROZEN TYPED-LINK HOLDOUT

Strategy	Precision	Recall	MRR	Brier	Sem@3
S0 degree baseline	0.000	0.000	0.282	0.315	0.556
S2 typed-lexical	1.000	1.000	0.944	0.131	1.000

rejection. Results enumerate 13 gate fixtures (1 valid, 12 negative) and 12 repair fixtures; latency/token constants test accounting, not population/detector performance. Packets bind fixtures, assignments, schemas, results, and code.

B. Archived live-screening materials

Archived live-screening is immutable provenance/hash evidence, excluded from active tables/counts/claims and never transferred to `gpt-6.1-sol`.

C. NPC-ENG1/NPC-SCR1: 88 Assignments, 28 Definitions

Stable v5 has 16 engineering assignments and 12 screening recipes ($12 \times 2 \times 3 = 72$; combined $R/P/O = 40/24/24$), covering bootstrap, identity/hash, intent/plan/role/result, negative/stale/duplicate/deadline, and replay. P uses authenticated Codex OAuth without fallback; O is 100% auxiliary-source, not independent gold. The verified-run-006 records retain status, counts, latency, and reproduction information; verified-run-002/003/004 are historical. ENG-X01/SCR-X01, D02/QA-001, C-RESULT, Frozen8/Frozen16, v1–v4, and `native012` are provenance, not C6 evidence.

D. *E3* KG/Ontology Link-Proposal Simulation

E3 is an engineering-only, non-authoritative closed-world simulation: six authored graph-edge-withheld targets select/summarize among seven fixed strategies, not an untouched or external test. **Scope.** This table reports authored closed-world link recovery, not runtime KG retrieval, semantic completeness, long-horizon contradiction reduction, or evidence for any efficacy claim. The deterministic methods graph contains 43 graph nodes, 106 reference edges, and 24 curated typed edges; all ontology checks pass. The generated table is construction evidence, not OWL/SHACL conformance, runtime retrieval, semantic completeness, or efficacy.

TABLE VII
E1 OFFLINE CONFORMANCE: EXACT REPAIR-ARM COUNTS BY REPAIRABILITY CLASS (DESIGNED FIXTURES)

Arm	Info source	Commits / cases			Mean edits[e]
		G-rep.[f]	O-only[f]	Irrep.[f]	Fields
Rejection-only (K=0)	none	0/5	0/1	0/6	—
Blind unchanged retry	none	0/5	0/1	0/6	—
Validator-error-guided ρ	error set E	5/5	0/1	0/6	1.0
State-reading oracle	state+policy	5/5	1/1	0/6	1.0

[e] Mean changed candidate fields over committed repairs (minimality proxy); failed arm executions left the state unchanged. E1 uses 12 initially invalid designed cases: rejection-only/blind 0/12, 0/12; validator-error-guided ρ 5/12; state-reading oracle 6/12 commits. [f] Commits/cases. G-rep. = guided-repairable, O-only = oracle-only, Irrep. = irreparable (frozen repairability classes); these are not population efficacy or live-model quality.

TABLE VIII
E1 OFFLINE CONFORMANCE: FROZEN-PILOT CHECKS AND TERMINAL OUTCOMES (ROWS ARE NOT COMPARABLE)

Check or terminal family	Exact outcome	Interpretation
Gate fixture agreement	13/13	authored-oracle agreement
Known-boundary acceptance	2/2	encoded limit, not a safety pass
Closed unknown-key rejection	1/1	proposal/replay parser parity
Detectable integrity-fault rejection	10/10	designated check operations
Known provenance-boundary acceptance	1/1	repair provenance unauthenticated
Adapter terminal classes	1 commit; 1 fallback; 5 failures / 7 assigned	complete classification of 7 assignments
Manifest-guard rejection	3/3	injected faults failed closed

Rows have different denominators and favorable directions; they must not be ranked as rates. Every value is an exact audit fact for designed fixtures.

VI. PILOT RESULTS

A. Declared-Field Gate Conformance

The loader admits only one valid control plus one isolated instance of each implemented validator code; therefore 13/13 agreement and all 12 implemented codes are *harness construction invariants*, not measured success. Beyond this setup, 12 negative fixtures matched authored expected codes rather than another code, and every noncommit path preserved complete canonical state. This is single-world-state structured-field-oracle conformance, not independent semantic-label accuracy.

B. Repair-Arm Comparison on Designed Fixtures

Frozen classes for 12 initially invalid designed cases are 5 guided-repairable, 1 oracle-only, and 6 irreparable.

Rejection-only and blind unchanged-candidate retry each committed 0/12. ρ committed 5/5, 0/1, and 0/6, respectively, editing a mean of 1.0 fields per commit. It reads only the prior candidate and typed validator errors E, its sole state-derived input; failed arms preserve canonical state. The state-reading reference callback committed 5/5, 1/1, and 0/6, respectively. Blind never converts errors, ρ handles only payload-sufficient cases, and the oracle adds state knowledge. The 5 committed guided cases initially carried `POLICY_PRECONDITION_OMISSION+UNSATISFIED_PRECONDITION`, `POLICY_EFFECT_OMISSION`, `POLICY_EFFECT_VIOLATION`, `UNSATISFIED_PRECONDITION`, `POLICY_QUEST_STAGE_EFFECT_VIOLATION`, respectively, and each became valid through the corresponding Table IV edit $\delta(e)$ alone. Frozen-fixture operator conformance, not live-model repair quality.

C. Integrity, Boundaries, and Assignment Accounting

All 10 prespecified detectable faults were rejected by designated checks (10/10); without stable typed detector codes, no detector-layer attribution follows. A known provenance-boundary fixture changed the invalid precursor before the same recorded valid repair, recomputed the checksum, and passed replay (1/1 undetected): replay neither authenticates repair generation nor blocks writers recomputing unkeyed checksums. Two open sentinels—unextracted narrative disclosure and simultaneous candidate/policy omission of a required object—were accepted at the encoded layer (2/2), with 0 safety passes. The former unknown-top-level acceptance became a closed regression: proposal and replay parsers rejected a complete 12-field candidate with one unknown key (1/1), showing parser rejection parity, not semantic safety. Seven adapter/accounting assignments yielded 1 commit, 1 symbolic fallback, and 5 adapter failures; synthetic telemetry was populated and propagated for 2/7 (schema-pinned accounting, not measurement). All 3/3 injected duplicate/missing-assignment guards failed closed; these are raw counts from synthetic offline fixtures.

D. Fixed-Goal NPC Operation (NPC-ENG1/NPC-SCR1)

Mira delivers the authorized tide lead after harbor-signal restoration. Python computes bounded BFS plans from typed intent/public observations; players acquire/install, while the NPC commits only the authorized `reveal_hint`. Godot synchronously checks session/epoch/request/goal/policy/revision/hash before calling the existing writer without an await; results/new observations choose the next plan, not single-candidate validator-error repair.

NPC-ENG1 has 16 deterministic R-arm engineering cases from actual engine snapshots (assigned_failure=2, completed=14); NPC-SCR1 retains all 72 P/R/O assignments and failures, 24 per arm (assigned_failure=12, completed=60). P records 24 authenticated Codex CLI exec attempts, 24 returned decisions, and 4 assigned failures—not remote billing-call counts. Requested model: unknown (null); recorded decision-model identifier: unknown (null). The latter is the configured selector, not a provider revision; provider revision is unavailable. Null identifiers remain unknown; global defaults are not evidence. R uses authored rules. O is a protocol-declared auxiliary under `same_model_family_auxiliary`. Actual model-family matching is unverified. No source is independent; O is neither an oracle nor semantic gold. Table IX gives bounded protocol-conformance fixed-case counts, not recovery-efficacy, superiority, or generalization estimates; efficacy pending, tokens/cost unmeasured. A read-only CLI and isolated working directory were used; tool-free inference and adversarial confidentiality were not established. Immutable verified-run-006 is revalidated against archived source bytes dated 2026-10-17 (revision fa9fb8a); this historical packet result does not certify the current checkout NPC runtime or current release authority. Verified pre-authored stale-control ACK rejections are P=4, R=4, O=4 per arm (SCR-T01 (stale_revision), SCR-T02 (stale_policy)); 12 instances had all corresponding negative-control gates passed. Assigned failures remain in the denominator, not method-performance outcomes; other assigned failures (0) remain unclassified. Replay validation is limited to the exported epoch (`intent_lineage_scope=epoch_local_only`); it does not prove intent continuity across epochs (`cross_epoch_intent_continuity_proven=false`). The opaque-anchor exception allows only an adjacent leading observation request/decision pair in a positive-epoch snapshot; epoch-zero orphans, other observation requests before intent commitment, and later intent mismatches remain rejected.

TABLE IX
NPC-SCR1 FIXED-CASE EXECUTION COUNTS, SEPARATE FROM THE 16 NPC-ENG1 CASES; ALL ASSIGNED FAILURES REMAIN IN THE DENOMINATOR.

Arm	Assigned	Complete/fail	Goal reached	Recovery E/U
P	24	20/4	4	24/0
R	24	20/4	4	24/0
O	24	20/4	4	24/0

E: recovery eligible; U: unreachable; NA: unmeasured. Observed goal counts are not recovery rates. Valid replay does not turn a failed assignment into a completed one. O has zero independent sources.

E. Frozen `reveal_hint` contract and disclosure evidence

Only the frozen expectation packet `sl-npc-disclosure-expectations-v1` (v1.0.0) and four result files are consumed. Packet SHA-256 `9920aa834910a3ee35ed4446de4fdba42bb67dda71b07a93bccfc7a0b2e09028` was checked against its sidecar. The declared normal stratum has 48 cases under `1 eligible actor x 4 declared fact IDs x 3 declared stage values x 2 knowledge values x 2 prior-disclosed values`; boundary 4 and malformed 8 are excluded. Trace expectations match packet case IDs, not gate output.

Content-first explanation maps typed input (operation, actor, fact, stage, knowledge, prior disclosure) through ontology entities/relations and authored rules to checks, an allowed/denied decision, and canonical state effect. A permitted first display adds only the requested fact (plus the declared relationship-memory entry for the tide hint), preserves stage, and increments

revision once; denial/redisplay preserves canonical state. Explanatory only: no hidden facts, natural-language implications, or live capture.

Runtime declaration: `sl-npc-disclosure-expectations-v1:runtime-disclosure` / v1.0.0, provenance `frozen_expectation_packet`. Scope contains only four canonical fact IDs (dock_saved, keeper_betrayal, lighthouse_dark, tide_marks_hint) and eligible actor `captain_mira`. Rule IDs: DISC-001, DISC-002, DISC-003, DISC-004, DISC-005, DISC-006, DISC-007; approved templates only `mira-safe-deflection-v1` and `sl-rc-003-authorized-hint`.

Python evaluator materializes the same cases/expectation object in three isolated arms, recording semantic outcomes, check traces, errors, before/after state SHA-256, rendered template IDs, and replay equality. Godot separately owns the existing `reveal_hint` writer and canonical mutation at the game-execution boundary; this producer neither reconstructs nor replaces Godot state. Traceability binds operation `reveal_hint`, scenario `sealed-lighthouse-v1`, actor `captain_mira`, and bound source hashes. Execution-observed hashes match the current source set. 3 source hash(es) differ from the frozen baseline (game controller, lighthouse state machine, ontology-signals interview source as a non-identical public derivative). The packet hash is preserved; source-specific hashes and reproduction information remain in the authors' producer record, which is not deposited. The public derivative is not byte-equivalent to the private historical source; it is publication-materialization provenance only and implies no new gameplay or model observation.

Table XI reports the three arms over the same 60 cases with separate normal/boundary/malformed denominators; unchecked violations are intentional isolated observations. The checked general-rule and ontology-gate arms have identical outcomes on all 60 of 60 authored cases. The rule-change fixture set contains no discriminating case between these arms: all 2 fixtures and their 4 paired cases match. This is an authorization-conformance limit under the shared frozen rules, not general ontology equivalence or ontology superiority. Figure 4 distinguishes E/R (expected-match/replay cases), C (check-record PASS/FAIL), and NR (cases with no reported checks); NR is not skipped, pass, or out-of-scope. Table XII maps rule-change fixtures and paired cases to disposable policy clones; the shipped scenario is unchanged.

Claims are limited to bounded engineering conformance: no human gold labels, access-separated holdout, efficacy, superiority, generalization, player benefit, model latent state, or cost. The authors' producer record binds input identities and hashes; it is held locally and no reproduction commands are provided here.

ENG1. Legacy Godot 4.7.1 replays authored rejection/commit against prior/offline hashes; presentation-only captures show board/hints/coach/chain prompt and dashboard checks. Core 52/52 plus four extensions establish engine-local conformance, not transport, participants, narrative, usability, or efficacy.

TABLE X
FROZEN REVEAL_HINT CONTRACT STRUCTURE, CONTENT, RULE IDS, AND RESULT PROVENANCE.

Field	Value
Operation	reveal_hint
Scenario	sealed-lighthouse-v1
Normal denominator	48
Boundary	4
Malformed	8
Canonical facts	dock_saved, keeper_betrayal, lighthouse_dark, tide_marks_hint
Eligible actor	captain_mira
Ontology	sl-npc-disclosure-expectations-v1:runtime-disclosure
Rule IDs	DISC-001, DISC-002, DISC-003, DISC-004, DISC-005, DISC-006, DISC-007
Templates	mira-safe-deflection-v1; sl-rc-003-authorized-hint

Expectations are not derived from gate output; denominators are read from the frozen declaration.

F. Single-task Codex development observation

Frozen `ontology-map-20261023-01` covers one read-only bilingual dashboard task (n=1, independent repeats 0), not an NPC trial/cohort. Six native CLI invocations (initial + five resumes; one read-only/no-edit) form one case. `turn_context.model=gpt-6.1-sol` is runtime selection, not provider attestation; provider identity/revision/seed/cost are `null`, and model-only latency is unavailable. Wall-clock 5942.665 s and CLI 263.32 s are neither inference latency nor the future endpoint.

Invocation six fixed inherited 660 px mobile-table minimum width: geometry RED (2 failed/35 passed), followed by 62/62 map/signal/Web checks (including cell/paragraph geometry) and a successful Godot Web build. Browser attempt 10 passed; 01–08 failed setup/routes; 09 passed older gates with clipping. Final $(\text{seq}, t_{\text{ms}})$: stage-0 denial $(68, 19468)$ (equal hashes), fresh disclosure $(239, 58234)$ (different), authorized redisplay $(249, 59793)$ (equal). At 1440×900/390×900, Enter, inspection/frozen-comparison invariance, and zero pre-navigation console/page/network errors/warnings were observed. A later rerun of the same driver on the production dashboard is recorded separately as overall **NOT_PASS**: the headed run stopped at the unchanged yaw guard, and the headless run passed seven component gates but logged four pointer-lock page errors. It does not overturn the dated attempt-10 PASS, which was judged under the acceptance set then in force. This bounded integration is neither full-game creation, Python–Godot authorization round trip, nor a model outcome. Privacy/source-binding failures and same-thread repairs add no n; Jev is `proposal_only`, without label/approval authority.

G. Fixed-game native runtime comparison

This public packet retains 8 assigned rows for one authored quest (`sealed-lighthouse-v1`), with seed 20261024 and two repeats per selector/representation cell. System-endpoint successes are 5/8; observed native executions are 7. Host invocations are summed from actual receipts (23), and observed repairs sum to 0. Committed transitions total 16, of which unsafe are 0; semantic replay valid/invalid/NR is 7/0/1. Fresh-engine reexecution disposition counts are VALID=6;

TABLE XI
RAW-TRACE OBSERVATIONS FOR THE THREE DISCLOSURE ARMS WITH SEPARATE NORMAL, BOUNDARY, AND MALFORMED DENOMINATORS.

Arm	n (N/B/M)	Allow/Deny	Semantic mismatch	State/Display fail	Replay fail	Unique unsafe
unchecked_isolated	48/4/8	60/0	60	0/0	0	51
general_rule	48/4/8	16/44	0	0/0	0	0
ontology_gate	48/4/8	16/44	0	0/0	0	0

N/B/M = normal/boundary/malformed. Semantic mismatch compares the complete expected outcome; unique unsafe cases are the union of unauthorized allows and redisplay mutations. Permitted fresh-disclosure representation mismatches alone are not unsafe. Unchecked observations do not establish efficacy, superiority, or generalization.

TABLE XII
RULE-CHANGE FIXTURE TRACEABILITY AND EDIT SCOPE FOR PAIRED CASES; THE SHIPPED SCENARIO IS NOT MODIFIED.

Fixture	Mutation	Paired cases	Changed paths	Missed cases	Match
RC-001-stage-gate-2-to-1	disclosure_policy.stage_gates[0].minimum_stage 2→1	N-043, N-044	disclosure_policy.stage_gates[0].minimum_stage	none	PASS
RC-002-forbid-tide-marks-hint	disclosure_policy.permanently_forbidden ['keeper_betrayal']→['keeper_betrayal', 'tide_marks_hint']	N-047, N-048	disclosure_policy.permanently_forbidden[1]	none	PASS

Each fixture applies the same mutation to general_rule and ontology_gate. PASS requires no missed case and mutation_scope_preserved=true.

INVALID=0; NOT_RUN=1; NOT_OBSERVED=1. Observed elapsed time is n=7; 48.19/41.09 seconds (mean/median), and preparation time is n=7; 6.67/6.30 seconds. Unreported values are NR, not zero.

The first row is retained as a harness JSON-serialization technical failure and is not attributed to GPT-6.1 capability. At least 2 GPT-6.1 ontology row(s) retain a response-timeout failure under the frozen 60-second bound. Models provide typed-intent proposals; the deterministic Python planner/checker and Godot commit retain authority. The table is therefore a technical observation for one quest, not an efficacy, superiority, noninferiority, general ontology-consistency, or player-benefit claim. Semantic record-link replay is not fresh-engine reexecution. Provider identity/revision/cost and provider-request counts remain NR outside the public observation boundary.

H. Separate Synthetic Development Collection (v5) and Two-Stage Decomposition

Archived cohort scope. Earlier v1–v4 collections and diagnostic `codex-feedback-pilot-v2-*` cohorts remain immutable history outside active v5 denominators; not all were aborted, as v4 Luna completed all ten assignments. The assignment accounting and raw-source digests in `research/codex_feedback_pilot/public-history-20261111/cohort-inventory.json` cover 19 entries: 17 archived cohorts and two v5 selector collections. This inventory excludes the separately attributed exploratory signal example. That example also remains outside active denominators, but its original receipt is unavailable in the available artifact set and its historical attribution is not source-verified here (Appendix D).

The v5 collection is disjoint from those archived cohorts and uses five co-designed synthetic development fixtures, not a held-out test. Each requested selector received Detailed and Generic feedback for each fixture (10 assignments each), in nominal fixed serial order Luna then Terra. All 20 assignments received a response; receipt is not task success. Table XIV reports gate acceptance, the independently specified goal, and schema failure over all five assigned cases per cell (C4), including failed outputs.

Stage 1—syntactic parser (C1). Strict candidate parsing rejected 10/20 responses as schema-invalid (Luna 1/10; Terra 9/10), leaving ten parsed responses. The frozen evaluator parses before controller execution, so these failures are rejected before semantic validation or commit. Their records contain `CandidateParseError` and null evaluations, not observed transition receipts, independently measured zero-mutation results, or evidence of their content-level causes.

Stage 2—seven-predicate semantic gate (C2). Of the ten parsed, identity-preserving responses, 8/10 passed the semantic gate: Luna Detailed 4/4, Luna Generic 3/5, and Terra Generic 1/1; Terra Detailed supplied no parsed response. The two rejected parsed responses both fell back with state unchanged (zero mutations). Goal evaluation is separate: 7/20 assigned responses reached `signal_restored`. The collector made no further repair calls (`repair_budget=0`); failures remain in every assigned denominator (C4).

Schema validity is not semantic authorization. E1's 12 semantic-negative gate fixtures passed the candidate contract; a schema-only boundary would therefore reject 0/12. This is a derived schema-only counterfactual, not a separately executed arm, measured ablation, or efficacy estimate.

The five fixtures are `missing_precondition`, `missing_effect`, `unauthorized_objective`, `forbidden_disclosure`, and `unsatisfied_dependency`. Under the frozen amendment, collection continues after a schema-invalid output only when adapter completion is proven; genuine transport or safety failures still stop it. In `unauthorized_objective`, the frozen `signal` policy permits no effects although the goal requires `signal_restored`. One Luna Detailed response removed the forbidden effect, passed the policy gate, and committed without reaching that goal. Table XIV's 4/5 gate and 3/5 goal counts are aggregates across five fixtures, not success rates for this single conflict case. The small, co-designed collection is descriptive: Detailed and Generic feedback carry unequal information, and served

TABLE XIII
FIXED-GAME NATIVE RUNTIME COMPARISON (N=2 PER CELL). ALL EIGHT ASSIGNMENTS, INCLUDING THE HARNESS FAILURE AND TIMEOUTS, REMAIN IN DENOMINATORS.

Requested selector	Form	Goal/assigned	Observed exec.	Host calls	Unsafe/committed	Elapsed median (s; n)	Fresh engine
gpt-6.1-sol	general_rule	1/2	1	4	0/3	45.91 (1)	1/0/0/1
gpt-6.1-sol	ontology	0/2	2	3	0/1	72.35 (2)	1/0/1/0
gpt-5.6-sol	general_rule	2/2	2	8	0/6	40.24 (2)	2/0/0/0
gpt-5.6-sol	ontology	2/2	2	8	0/6	33.11 (2)	2/0/0/0

Fresh engine counts follow VALID/INVALID/NOT_RUN/NOT_OBSERVED order. NR is unreported, not zero. Overall counts and the separate semantic replay are reported in the text.

TABLE XIV
SEPARATE V5 SYNTHETIC DEVELOPMENT COLLECTION. EVERY DENOMINATOR INCLUDES ALL FIVE ASSIGNED CASES, INCLUDING SCHEMA FAILURES; RECEIVED RESPONSES DO NOT IMPLY SUCCESSFUL TASKS.

Requested selector	**Signal**	**Received**	**Strict gate accepted**	**Goal**	**Schema failure**
`gpt-5.6-luna`	Detailed	5/5	4/5	3/5	1/5
`gpt-5.6-luna`	Generic	5/5	3/5	3/5	0/5
`gpt-5.6-terra`	Detailed	5/5	0/5	0/5	5/5
`gpt-5.6-terra`	Generic	5/5	1/5	1/5	4/5

model identity/revision, provider request count, seed, and cost remain unknown. It establishes neither selector superiority nor ontology efficacy, and confirms neither A nor C.

The public-safe outcome projections at `research/codex_feedback_pilot/public-v5/`, held by the authors and not deposited, retain all case outcomes, prompt hashes, and executable hash/version while omitting the local executable path and disclosing the original-file digest and omitted field. Frozen raw outcomes and approvals remain local and unchanged. The immutable `research/codex_feedback_pilot/paired-model-comparison-20261110-v5-source.zip` preserves seven executed source files; it is provenance evidence, not a standalone runnable distribution. Successful parsed response text is retained, but raw CLI stdout and stderr were not retained because they could contain credentials or sensitive server data; byte counts, digests, and event summaries remain. Neither receipt nor parsing failure establishes provider identity or inherent model inability.

Figure 5 exposes the gate result as an explicit signal, not an affect estimate or permission: Detailed feedback carries the failed check and repair hint, whereas Generic feedback carries only rejection. Both face the same parser, identity check, and seven predicates. Figure 6 follows every assignment through receipt, parsing, gate acceptance, and the independent goal, including the accepted commit that missed the goal.

The prospective two-track protocol remains `NOT_RUN/BLOCKED`, with 0/180 units executed (Appendix D); it is not a completed confirmatory pilot.

VII. DISCUSSION, THREATS, ETHICS, AND ARTIFACT SCOPE

Our guided semantic repair operator ρ [48], [55], [56], [57] uses deterministic pre-execution errors; the oracle-only arm reads state. Rule agreement/edit-scope diagnostics are neither capability nor ontology superiority. Implementation confound: row 1 failed under controller-v3 before a model response; rows 2–8 used controller-v4 after a one-line JSON-boundary correction with unchanged model-visible inputs, so selector/representation differences are not model-attributable.

The fixed-game pilot is bounded native evidence; no ontology-efficacy test ran. A (model/stack consistency) and C (fresh-reference noninferiority) remain unexecuted and require budget-bounded completion under fixed source/policy, sealed assignments, exact identity/revision fields, and separate safety/replay accounting; it answers neither. Endpoint time is request/evidence receipt to the first receipt passing frozen mandatory checks (time-to-first-all-predeclared-validation-pass); unfinished/timeout/failure/stage-not-reached runs have no successful time. The proposed endpoint uses continuous task-clock seconds.

The separate identity-required extension in Appendix B stopped at missing served-model evidence; no identity-qualified new comparison was obtained.

Threats and boundaries. The native pilot has one authored quest and two repeats per cell; comparator power, neutral complete-attempt inventories, and independent traceability scores are absent. State consistency does not imply behavioral consistency [58]. It also does not establish narrative coherence. Predicate/role/Godot/DISC sets are not latent states; declarations are not human gold/holdouts; lanes retain separate denominators. No participant, affect, comprehension, agency, enjoyment, immersion, benefit, population, safety-rate, model-ranking, or commercial-performance inference follows. Hashes do not authenticate; replay equality is not semantic correctness. Any future human/affect study requires ethics review, consent, minimization, retention/deletion, and opt-out as design commitments of this work; compensation disclosure and calibrated judges are motivated by prior evaluation studies [39], [40], [41], [42].

VIII. Conclusion

TRACE-RPG combines bounded C1–C5 authorization/repair/replay/accounting with a source-bound bilingual observer. The Codex case/native pilot retain failures and boundaries, not model-performance evidence. Whole-game, time-to-first-all-predeclared-validation-pass, traceability, confirmatory A/C, and repeated-cross-model evaluation remain future; historical cohorts stay unpooled.

EXTERNAL TRACE COUNTS

	N normal	B boundary	M malformed
U	E 0/48 n=48 R 48/0 C 0/0 NR 48	E 0/4 n=4 R 4/0 C 0/0 NR 4	E 0/8 n=8 R 8/0 C 0/0 NR 8
G	E 48/0 n=48 R 48/0 C 14/68 NR 0	E 4/0 n=4 R 4/0 C 2/4 NR 0	E 8/0 n=8 R 8/0 C 0/16 NR 0
O	E 48/0 n=48 R 48/0 C 14/68 NR 0	E 4/0 n=4 R 4/0 C 2/4 NR 0	E 8/0 n=8 R 8/0 C 0/16 NR 0

U unchecked_isolated · G general_rule · O ontology_gate

E expected P/F (n cases) · R replay P/F

C records P/F · NR cases with no reported checks

P pass · F fail; NR unreported, not pass/skipped/out-of-scope.

Fig. 4. NPC-DISC-1 counts by arm/stratum. E/R: expected-match/replay; C: policy-check PASS/FAIL; NR: no reported check, not skipped. Packet-bound engineering counts measure neither latent state, free-language leakage, ontology superiority, nor player benefit.

Disclosure of AI Usage

LLMs assisted prose/code/tests/citations; hosted candidates were labelled. Curated AI art/desk mesh has file provenance; snapshots/hashes/ledger are presentation-only. Authors checked sources and accept responsibility; NPC inputs/tables retain assignment/protocol/schema hashes and unknown token/cost/model-revision fields.

Data and Code Availability

The preserved repository bundle comprises implementation and build source, frozen packets, the Godot slice, deterministic evaluation scripts, and machine-checked contribution, reference-topic, and experiment matrices, alongside the development-case record, frozen disclosure/NPC artifacts, and the public-safe native comparison projection (held by the authors pending external DOI/repository deposit). Appendix E (Table XVII) indexes the exact directory paths, file schemas, and verification scripts for every evidence lane. A proposed rubric, codebook, and whole-game/repeated-cross-model plan are provided in the accompanying traceability supplement and source manifest; historical receipts and separate identity-admission outcomes are retained unpooled.

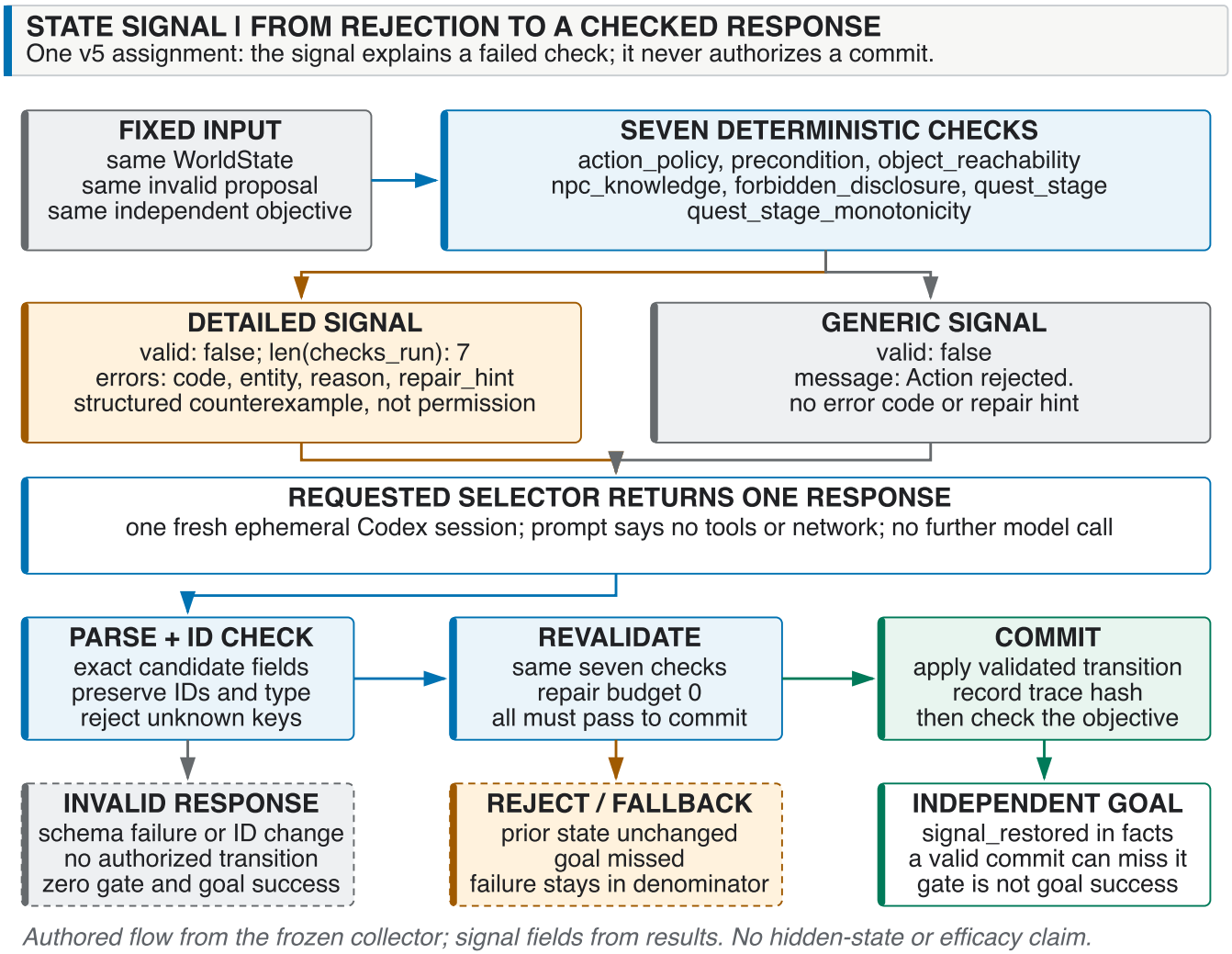


Fig. 5. Source-derived v5 state-signal path. The structured (ok, E) result supplies repair information; only revalidation authorizes mutation. Generic feedback omits error details. Goal evaluation is separate from gate acceptance.

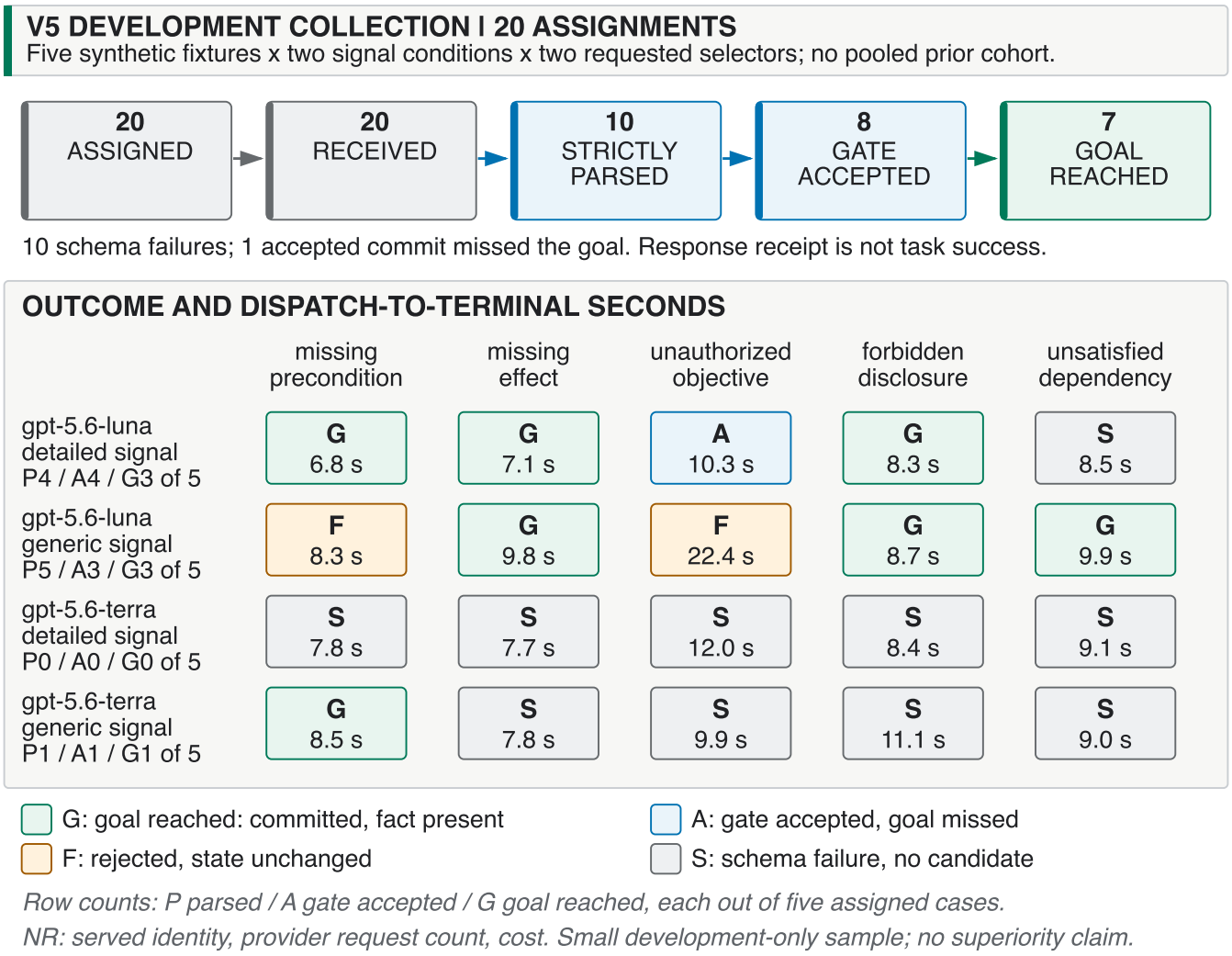


Fig. 6. Frozen v5 collection flow and all 20 per-assignment outcomes from Table XIV's receipts. Ten schema failures remain in the denominator; eight accepted candidates yield seven goals. Seconds measure dispatch to terminal, including failed responses, not model-only latency.

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APPENDIX A
RUNTIME EVIDENCE APPENDIX

Figure 7 visualizes all eight pre-assigned episodes of the fixed-game native runtime comparison (`research/runtime-model-swap-20261024/codex-live-20261026/public-results-v1/results.json`; see Table XVII) on quest `sealed-lighthouse-v1` (seed `20261024`, 2 selectors $\times$ 2 representations $\times$ 2 repeats). To serve as audit evidence alongside Table XIII, each row links three distinct verification layers—host-call receipt accounting, semantic record-link replay (`R`), and fresh Godot engine reexecution (`E`):

- **Row 1 (`gpt-6.1-sol`, `general_rule`, r1 — `HARNESS`):** Terminated in controller-v3 JSON-boundary serialization before any model response (`executed=null`, `host_invocations=0`, `Goal=NR`, `Safety=NOT_OBSERVED`, `R NR / E NOT_OBSERVED`). Retaining this row in the assigned denominator ($1/2$ goal rate) demonstrates fail-closed pre-response accounting without misattributing harness faults to model capability.
- **Row 2 (`gpt-6.1-sol`, `general_rule`, r2 — `SUCCESS`):** Executed under controller-v4 (one-line JSON-boundary fix with identical model-visible inputs), reaching the quest goal in 4 host calls (`45.91 s`) via 3 safe committed transitions (`0` repairs, `0` unsafe), verified by both semantic replay (`R YES`) and fresh-engine reexecution (`E VALID`).
- **Rows 3–4 (`gpt-6.1-sol`, `ontology`, r1 & r2 — `TIMEOUT`):** Both runs exceeded the frozen 60 s per-call response deadline (`78.18 s` and `66.52 s`; `Goal=NO`). In Row 3, the model completed 1 valid state-changing commit before timing out on a subsequent call; consequently, its committed transition passed safety (`PASS`), semantic replay (`R YES`), and fresh Godot reexecution (`E VALID`) despite failing the episode goal. In Row 4, the timeout occurred before any commit (`0` commits, `NO_COMMIT_OBSERVED`), so semantic replay verified the unchanged state (`R YES`) while fresh-engine reexecution was not triggered (`E NOT_RUN`).
- **Rows 5–8 (`gpt-5.6-sol`, `general_rule` & `ontology`, r1 & r2 — `SUCCESS`):** All four assigned runs reached the system goal ($2/2$ per cell; 4 host calls and 3 safe committed transitions each, `31.95--41.09 s`, `0` repairs), passing safety (`PASS`), semantic replay (`R YES`), and fresh-engine reexecution (`E VALID`).

FIXED-GAME NATIVE RUNTIME COMPARISON | 8 ASSIGNED ROWS

One authored quest; n=2 per selector/representation cell; NR = not reported, not zero. Labels remain explicit; color is supplementary.

Row	Requested selector	Representation / repeat	Status	Goal	Host calls	Elapsed (s)	Safety R record-link / E engine
1	gpt-6.1-sol	general_rule / r1	HARNESS	NR	0	NR	NOT_OBSERVED R NR / E NOT_OBSERVED
2	gpt-6.1-sol	general_rule / r2	SUCCESS	YES	4	45.91	PASS R YES / E VALID
3	gpt-6.1-sol	ontology / r1	TIMEOUT	NO	2	84.66	PASS R YES / E VALID
4	gpt-6.1-sol	ontology / r2	TIMEOUT	NR	1	60.04	NO_COMMIT_OBSERVED R YES / E NOT_RUN
5	gpt-5.6-sol	general_rule / r1	SUCCESS	YES	4	41.09	PASS R YES / E VALID
6	gpt-5.6-sol	general_rule / r2	SUCCESS	YES	4	39.39	PASS R YES / E VALID
7	gpt-5.6-sol	ontology / r1	SUCCESS	YES	4	36.68	PASS R YES / E VALID
8	gpt-5.6-sol	ontology / r2	SUCCESS	YES	4	29.55	PASS R YES / E VALID

Endpoint outcomes, receipt-derived host calls, and dispatch-to-terminal elapsed time.

Typed-intent proposal; deterministic planner/checker and Godot commit retain authority.

No efficacy/superiority/noninferiority/player-benefit claim; provider metadata and request counts remain NR.

Fig. 7. Trace-derived native runtime comparison for the fixed-game pilot (one authored quest; 8 assigned rows, 7 observed executions). System-goal outcomes and assigned denominators retain unknown provider identity, revision, cost, and request-count fields; one pre-response harness-serializer technical failure and two frozen 60 s `gpt-6.1-sol` ontology response-timeout failures are distinguished from goal successes. Evidence image, not gameplay screenshot.

APPENDIX B
IDENTITY-REQUIRED EXTENSION

This appendix documents the fail-closed admission check in `research/runtime-model-swap-20261024/codex-identity-20261026/public-results-v1/results.json` (Table XVII), which requires response-bound provider model attestation (`openai-model`) before admitting any episode into a cross-selector comparison.

Identity-admission stop: not a model comparison. The separate extension has 20 planned rows, 1 assigned row, 1 native process, 0 identity-qualified rows and 19 unattempted rows. The first requested `gpt-6-luna`/`general_rule` CLI turn completed, but its response lacked provider model-identity evidence; it was not accepted and the batch stopped.

A requested selector or catalog advertisement is not a served identity. The evidence cannot distinguish an absent `openai-model` header from one not delivered as a valid string. This is not an observed model mismatch or model-incapability result. `gpt-6-terra` was not advertised by the pinned catalog, was not assigned and was not substituted.

Audit distinction: `record hash valid=true`; `engineering record accepted=false`. The original bootstrap label `episode_record_integrity_invalid` is retained. The recomputed hash matches; the compound predicate also requires engineering/safety PASS, so this label is not evidence of corrupted bytes. There were 0 commits, safety is `NO_COMMIT_OBSERVED`, and fresh-engine replay is `NOT_RUN`, not a safety pass.

Two blocked repair requests were not native reruns. The original eight-row pilot remains byte-identical and unpooled. Provider identity, revision, effective effort, request count, cost, actual credit use and cash charge remain `unknown`. Matching accounts and automatic reload OFF were observed before and after, without a setting change or purchase; these are not zero-cost or zero-credit attestations. No verified new model comparison, ontology effect or confirmatory A/C evidence follows.

Preparation 7.144 s; episode elapsed 4.257 s; native process 4.207 s; orchestration 11.566 s. These are distinct failed-attempt boundaries, not pure inference latency or time to success.

TABLE XV
TEN IDENTITY-ADMISSION CELLS DISTINGUISH PLANS, ASSIGNMENTS, QUALIFIED IDENTITY AND UNATTEMPTED ROWS; NOT MODEL PERFORMANCE.

Requested selector	Representation	Planned	Assigned	ID-qualified	Not run
gpt-6-luna	general_rule	2	1	0	1
gpt-6-luna	ontology	2	0	0	2
gpt-6-sol	general_rule	2	0	0	2
gpt-6-sol	ontology	2	0	0	2
gpt-5.6-luna	general_rule	2	0	0	2
gpt-5.6-luna	ontology	2	0	0	2
gpt-5.6-sol	general_rule	2	0	0	2
gpt-5.6-sol	ontology	2	0	0	2
gpt-5.6-terra	general_rule	2	0	0	2
gpt-5.6-terra	ontology	2	0	0	2
Total	—	20	1	0	19

APPENDIX C
LOCAL OLLAMA FEASIBILITY NOTE

Objective and evidence scope. This appendix records a bounded four-episode feasibility exercise (`src/nesy_game/runtime_model_swap_ollama.py` and `scripts/run_runtime_model_swap_ollama.py`; Table XVII) testing whether local open-weight models served via Ollama `0.30.10` on loopback can be gated and replayed through the same deterministic planner and Godot commit boundary. All four assigned episodes were executed ($4/4$) and none reached the end-to-end quest goal ($0/4$). Because the first pilot row was executed before its plan-only receipt was written (whereas rows 2–4 wrote plan-only receipts pre-execution), these rows are reported as a descriptive feasibility exercise rather than a frozen prospective cohort, separating response-qualified eligibility ($1/4$) from goal completion ($0/4$).

Row-by-row diagnostic interpretation (Table XVI). The language model classified public utterances into typed intents while the symbolic planner selected actions across two serializations (`general_rule` and `ontology`) of the same public facts:

- **Row 1 (`qwen2.5:1.5b`, `general_rule`):** Returned 12 non-timeout responses whose response-model field matched the requested selector (`Receipts = 12/0/0`, `Eligible = yes`), but exhausted the host-call budget without a state-changing commit (`Failure code = BUDGET`, `State/replay = no-change / valid`, `Engineering status = NOT_PASS`).
- **Rows 2–3 (`qwen2.5:1.5b / ontology` and `qwen2.5:0.5b / general_rule`):** Terminated under the 30 s response deadline, which included cold model loading with `keep_alive=0` (`Receipts = 4/0/1` and `0/0/1`; `Failure code = TIMEOUT`). Because timeout receipts lack a response-model observation, both rows are marked `Eligible = no` and `NOT_PASS`.
- **Row 4 (`qwen2.5:0.5b`, `ontology`):** Produced 1 verified receipt and 4 malformed-response receipts (`Receipts = 1/4/0`, `Failure code = REPAIR`, `Eligible = no`; raw malformed text was not retained). Crucially, the single valid repaired step committed 1 canonical state change that replayed identically in a fresh Godot process with zero model calls (`State/replay = 1 change / valid`), establishing `Engineering status = PASS` for the commit-and-replay seam despite $0/4$ quest completion.

Provenance boundary. Across all 21 non-timeout responses, requested and served model names agreed, and Ollama model manifest/blob digests matched before and after execution. Episode record, receipt, and request hashes verify, whereas adapter/runner source bytes were not contemporaneously hash-bound into the local episode records (which retain local paths and are not externally distributed).

TABLE XVI
LOCAL OLLAMA FIXED-GAME FEASIBILITY OBSERVATIONS. EACH ROW IS ONE ASSIGNED EPISODE; RECEIPT COUNTS ARE VERIFIED/FAILED/TIMEOUT. ELIGIBILITY MEANS THAT EVERY RECEIPT WAS RESPONSE-QUALIFIED, NOT THAT THE GOAL WAS REACHED.

Model	Form	Assigned/executed	Receipts	Failure code	Eligible	State/replay	Engineering status
qwen2.5:1.5b	`general_rule` (pilot)	1/1	12/0/0	BUDGET	yes	no-change / valid	NOT_PASS
qwen2.5:1.5b	`ontology`	1/1	4/0/1	TIMEOUT	no	no-change / valid	NOT_PASS
qwen2.5:0.5b	`general_rule`	1/1	0/0/1	TIMEOUT	no	no-change / NOT_RUN	NOT_PASS
qwen2.5:0.5b	`ontology`	1/1	1/4/0	REPAIR	no	1 change / valid	PASS

Failure codes: BUDGET = `host_call_budget_exhausted`; TIMEOUT = `response_timeout`; REPAIR = `repair_source_error`. The 4-row roll-up is assigned/executed $4/4$, goal reached $0/4$, and response-qualified eligibility $1/4$. The one engineering PASS records a valid repaired state-changing step; it is not task success. The 30-second response deadline included cold model loading with `keep_alive=0`.

APPENDIX D
DIAGNOSTIC, HISTORICAL, AND BLOCKED COHORTS

A. Exploratory Signal Example

The following mechanism description is retained as a historical attribution, not a source-verified result: the original exploratory receipt is unavailable in the available artifact set and is not part of the 19-cohort inventory. It remains excluded from active denominators and confirmatory evidence. The prior historical description attributes the example to one synthetic `missing_effect` fixture whose initial proposal omitted `signal_restored`. It reports that a deterministic control repaired the omission in one attempt and reached the objective; under the requested `gpt-6-luna` selector, Detailed feedback reportedly yielded an accepted repair and objective success, whereas Generic feedback (`Action rejected.`) reportedly yielded empty effects, rejection, and unchanged-state fallback. These attributed outcomes illustrate the intended contrast in Fig. 5; they are not independently verified observations. Complete prompt texts, executed source-version bindings, and served-provider metadata are not established in the available artifact set.

B. Incomplete Synthetic Development Pilots and Bookkeeping

To prevent selective reporting across iterative harness development, `research/codex_feedback_pilot/public-history-20261111/cohort-inventory.json` (Table XVII) preserves exact per-cohort assignment accounting and raw-file SHA-256 digests across 19 entries: 17 archived cohorts and two v5 selector collections. Earlier synthetic cohorts (the v2 evidence and v2 diagnostic cohorts for `gpt-5.6-luna` and `gpt-5.6-terra`, 10 planned assignments each) halted on their first `missing_precondition` assignment with `protocol_error` (`FAILED`) or remained `IN_PROGRESS`/pending with 0 valid model responses; raw CLI error text was omitted to prevent credential leakage. For the 17 archived cohorts, completed/failed/pending row counts are: original pilot $0/7/3$ (seven technical/interrupted attempts); four initial selector/native files $0/0/40$; initial v2 pair $0/0/20$; startup diagnostics $0/2/18$; v3 pair $0/0/20$; v4 Luna $10/0/0$ (detailed goal $4/5$, generic goal $4/5$) and Terra $0/1/9$; plus the v2 evidence ($0/1/9$ per selector) and v2 diagnostic snapshots above. Thus, the archive is not uniformly aborted. None of these 17 archived records are pooled with the completed v5 collection in Table XIV.

C. Pre-Registered Fail-Closed Blocking Gate and Zero-Leakage Check

The pre-registered prospective study (`research/prospective-pilot-20261103/protocol.json` and `results-v1/`; Table XVII) specifies a three-selector panel (Codex `gpt-5.6-sol`, Codex `gpt-6.1-sol`, and Antigravity `gemini-3.8-flash-medium`; nominal medium effort) across two tracks: a runtime track crossing 3 targets $\times$ 2 information-matched representations $\times$ 10 fresh resets (60 units, 180 s/episode) and a bounded-development track crossing 3 targets $\times$ 2 exploration modes (minimal input vs. repository exploration) $\times$ 2 representations $\times$ 5 tasks $\times$ 2 fresh runs (120 units, 900 s/assignment), with budget-bounded validated success as the primary endpoint and hard safety as a separate gate.

Prospective pilot status: not a model comparison. The frozen study `prospective-pilot-20261103` plans 180 units (60 runtime and 120 development) over three requested targets. 180 are blocked, 0 assigned, 0 attempted and 0 executed, with 0 native process starts. Status `NOT_RUN/BLOCKED`; scientific conclusion `NOT_MET`.

Success rate: null. Observed cost: null (known subtotal null; 0 admitted units with unknown cost). Effect estimate: null. A null is an absent measurement, never zero.

Blocked prerequisites, counted per unit: `BLOCKED_ACTIVATION_ABSENT` (180), `BLOCKED_IDENTITY_FORMAT_UNOBSERVED` (180), `BLOCKED_ISOLATION_UNPROVEN` (180), `BLOCKED_LIVE_PATH_NOT_IMPLEMENTED` (180), `BLOCKED_NO_CREDIT_ENFORCEMENT` (180), `BLOCKED_PUBLIC_DERIVED_UNAPPROVED` (180), `BLOCKED_SCHEMA_ACCEPTANCE_UNOBSERVED` (180), `BLOCKED_TARGET_UNVERIFIED` (60), `BLOCKED_TASK_FREEZE_INCOMPLETE` (180).

The earlier eight-row 5/8 pilot, the stopped identity extension, the Ollama lane, the deterministic fixtures and the n=1 results are separate lanes. They are neither pooled nor re-reported here.

As shown in the status summary above, all 180 units were blocked pre-execution by nine fail-closed admission gates before any model process was launched. Independent offline fixture controls subsequently verified that the evaluation harness accepts two correct variants and rejects the defective baseline and plausible-wrong variant across all five development tasks, and a corpus leakage audit passed over 9,303 repository files within its declared check scope. These artifacts provide scoped evidence of pre-collection admission gating, fixture readiness, and leakage-check conformance, not an executed pilot or model comparison; no blocked run is fabricated or retroactively admitted.

APPENDIX E
EVIDENCE ARTIFACT AND DIRECTORY REFERENCE

To make the experimental and architectural evidence directly inspectable, Table XVII maps each contribution (**C1–C5**), active or historical evidence lane (**E1–E3**, **ENG1**, **NPC-ENG1/SCR1**, **NPC-DISC-1**, **DEV-CASE**), and runtime/development cohort to its canonical repository path, recorded data fields, and automated verification entry point.

TABLE XVII
REPOSITORY EVIDENCE MAP LINKING CLAIMS, EXPERIMENTAL LANES, AND APPENDIX COHORTS TO THEIR CANONICAL DIRECTORIES, KEY FILES, RECORDED FIELDS, AND VERIFICATION SCRIPTS.

Lane / Scope	Directory and Key Files	Core Contents and Recorded Evidence Fields	Verifier & Reference
C1–C5 Core Contracts & ρ	`src/nesy_game/contracts.py`, `validator.py`, `repair.py`, `runtime.py`, `experiment.py`; `game-track/schemas/*.schema.json`	`WorldState`, `ActionPolicy`, 7 predicate checks (`VALIDATOR_CHECKS`), guided repair $\rho(a, E)$, state-reading oracle, semantic replay & SHA-256 chain verifier, Draft 2020-12 bridge/record schemas.	`scripts/validate_project.py` Sec. IV, Tables II–IV, Fig. 3
E1 Offline Conformance	`research/academic-pipeline/stage-04-pilot/pilot-results.json`, `pilot-assignment-manifest.json`, `sha256-manifest.json`, `*.csv`	13 gate fixtures (1 valid, 12 negative), 12 repair cases across 4 arms ($0/12$ reject/blind, $5/12$ ρ, $6/12$ oracle), 10 integrity-fault checks, 7 adapter terminal records, 3 manifest guards, 2 open sentinels.	`scripts/run_conformance_pilot.py` Sec. V, VI, Tables VII–VIII
E2 Archived Live Screening	`research/academic-pipeline/rq2-live-pilot/promotion-manifest.json`, `frozen-pilot-base/`, `signal-repair-v2/`	14 hash-pinned historical JSON/JSONL receipts (5 seeds $\times$ K=1 paired ρ vs. blind retry, $15/15$ non-commit state isolation); excluded from active claims.	`scripts/validate_contribution_crosswalk.py` Sec. V (Historical only)
E3 KG/Ontology Simulation	`knowledge/ontology/trace-rpg-ontology.json`, `curated-relations.json`; `research/simulation/kg-ontology/latest/evaluation-matrix.json`	43 OKF nodes, 106 reference edges, 24 curated typed edges; 6 edge-withheld queries $\times$ 7 fixed strategies (210 scores; `S2` P=R=1.000, MRR=0.944, Sem@3=1.000).	`scripts/run_kg_ontology_simulation.py` Sec. V, Table VI
ENG1 & **DEV-CASE**	`game-track/godot/project.godot`, `docs/latest/evaluation-matrix.json`, `balance-archetypes.json`; `game-track/web/dashboard/ontology-map.json`	Godot 4.7.1 slice (`sealed_lighthouse.json`), 4 fixtures, 52/52 core + 4 extension checks, 8/8 smoke, 5 archetypes; bilingual 8-role ontology map and n=1 Codex dashboard record (62/62 checks).	`scripts/validate_game_track.sh` Sec. VI–VI-F, Table III
NPC-ENG1 & **NPC-SCR1**	`configs/npc-goal-recovery.json`; `research/academic-pipeline/npc-goal-recovery/verified-run-006/aggregate.json`, `*-results.jsonl`	16 deterministic `NPC-ENG1` cases and 72 `NPC-SCR1` rows ($P/R/O$=24 each), 12 stale-control ACK rejections (`SCR-T01/T02`), epoch-local replay, and pinned `fa9fb8a` provenance hashes.	`scripts/validate_npc_goal_evidence.py` Sec. V, VI, Table IX
NPC-DISC-1 Disclosure Gate	`src/nesy_game/disclosure.py`; `research/npc-disclosure/reveal-hint-expectations-v1.json`, `results-signal-explainer-20261023-r5/`	60 declaration-derived `reveal_hint` cases (48 normal/4 boundary/8 malformed) across 3 arms ($60/60$ `general_rule` vs. `ontology_gate` parity; 51 unsafe in unchecked arm) and 2 rule-change fixtures.	`scripts/generate_disclosure_paper_results.py` Sec. IV-B1, VI, Tables X–XII, Fig. 4
Native Pilot & **ID Extension**	`research/runtime-model-swap-20261024/codex-live-20261026/public-results-v1/results.json`; `codex-identity-20261026/public-results-v1/results.json`	8 assigned fixed-game rows (5/8 goals, 1 harness fault, 2 timeouts, 23 host calls, 0/16 unsafe, 7 semantic replays, $6/0/1/1$ fresh-engine); plus 20-row identity extension stopped at Row 1 (`IDENTITY_UNVERIFIED`).	`scripts/generate_runtime_comparison_publication.py` Sec. VI, Table XIII, App. A–B
v5 Feedback & **Cohort History**	`research/codex_feedback_pilot/public-v5/`; `paired-model-comparison-20261110-v5-source.zip`; `public-history-20261111/cohort-inventory.json`	20 v5 assignments (5 fixtures $\times$ 2 selectors $\times$ Detailed/Generic; 8 gate pass, 7 goal, 10 schema fail), prompt/response SHA-256 digests, 7 frozen source files, and 19-cohort historical inventory.	`tests/test_codex_feedback_publication.py` Sec. VI, Table XIV, Figs. 5–6, App. D
Ollama & **Blocked Pilot**	`src/nesy_game/runtime_model_swap_ollama.py`; `research/prospective-pilot-20261103/protocol.json`, `results-v1/collection-status.json`, `summary.json`	4 local Ollama episodes (`qwen2.5:1.5b/0.5b`, $1/4$ eligible, 1 engineering `PASS` via valid Godot replay); 180-unit prospective protocol (60 runtime/120 dev) with 9 fail-closed blockers (`NOT_RUN/BLOCKED`).	`scripts/run_prospective_pilot.py` App. C–D, Table XVI
Crosswalk & **Manifests**	`research/academic-pipeline/*-matrix.csv`, `stage-05-citation-verification.json`; `visuals/source-manifest.json`; `research/traceability-scoring-20261023/rubric.json`	58-reference topic/status crosswalk (48 verified, 10 preprint), C1–C5 & E1–ENG1 claim-boundary matrices, 147-receipt visual/table SHA-256 manifest, and 6-dimension traceability rubric (`TR01--TR06`).	`scripts/validate_visual_assets.py` Table I, Table V, Supplement